Kernix 0.7.1 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

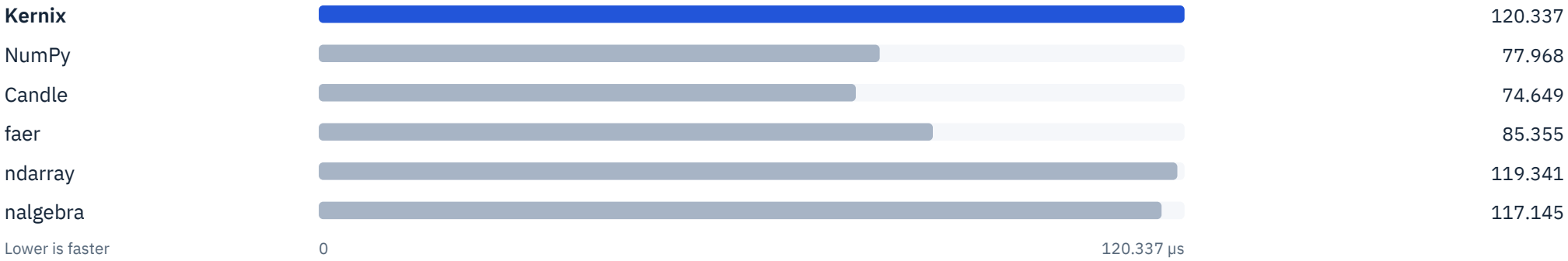
16 workloads

62 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

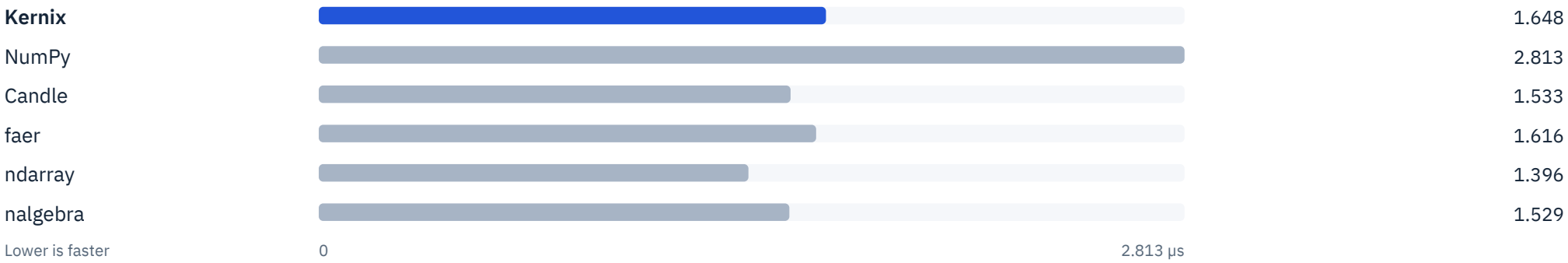
Kernix 0.7.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

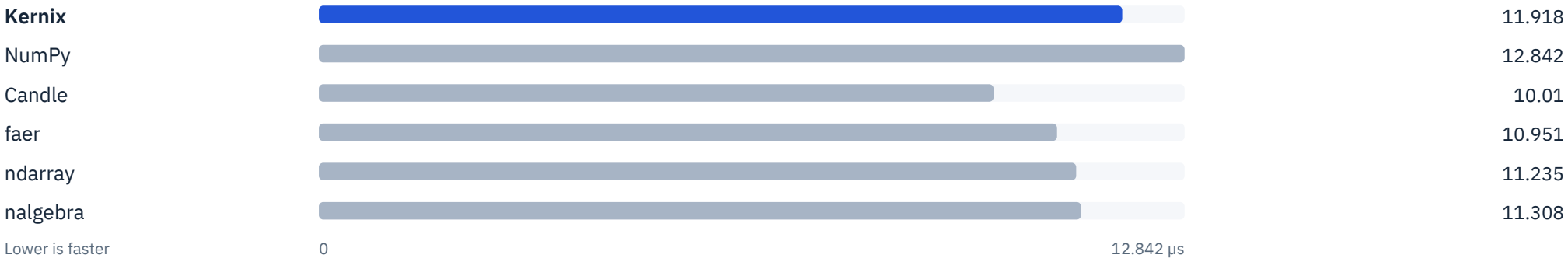
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



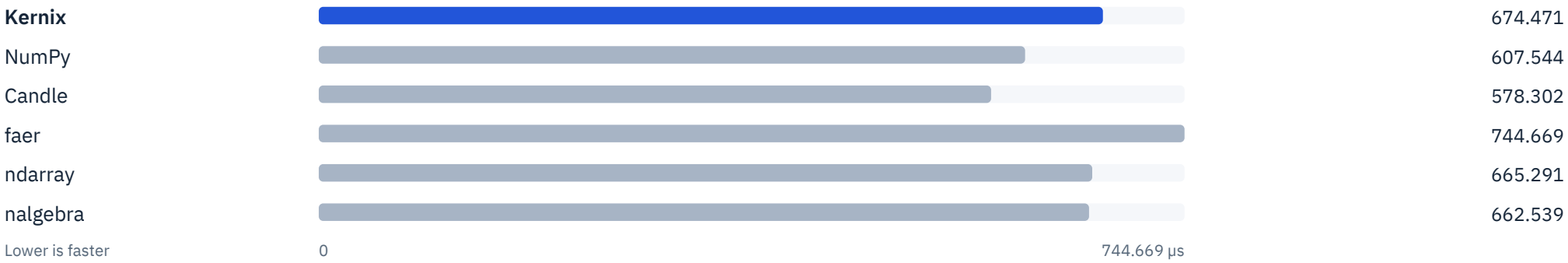
Kernix 0.7.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

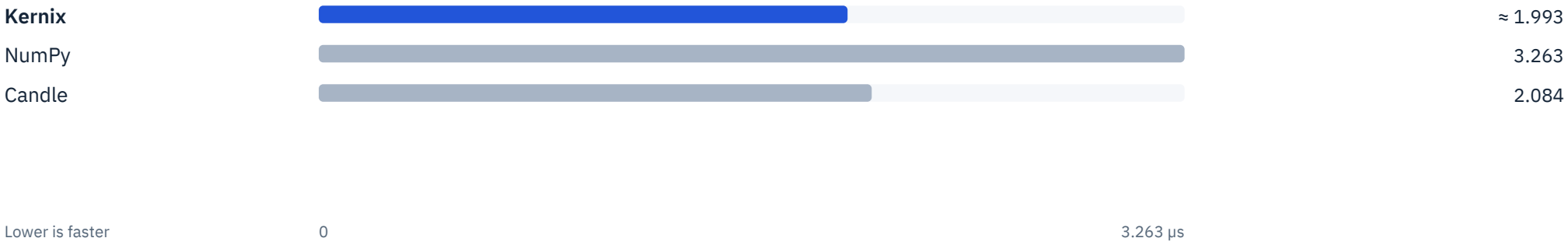
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



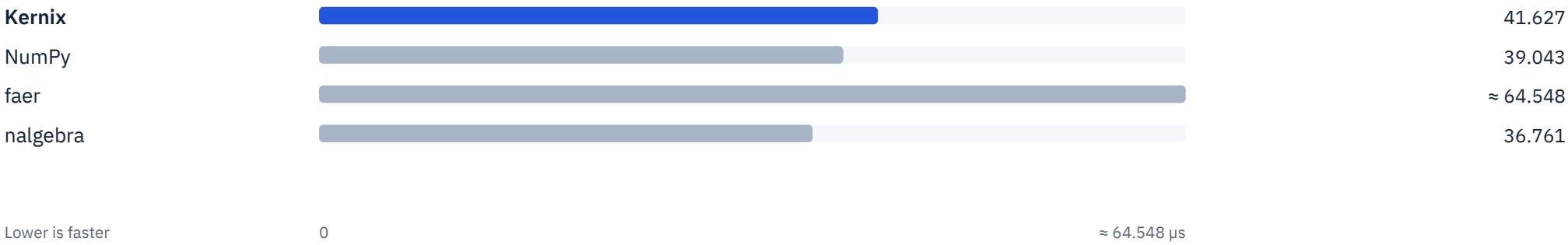
Kernix 0.7.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

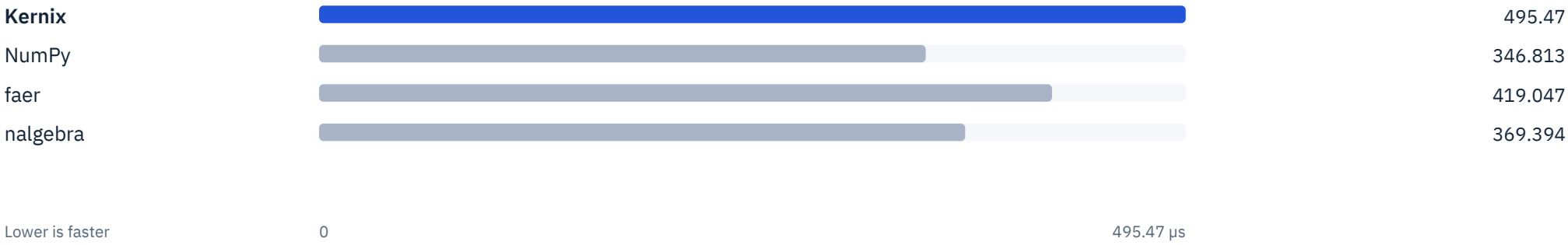
Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median

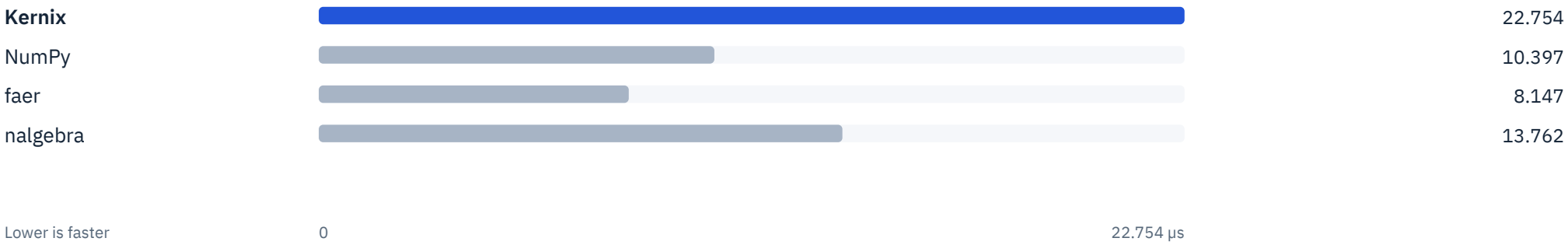


Kernix 0.7.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.7.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

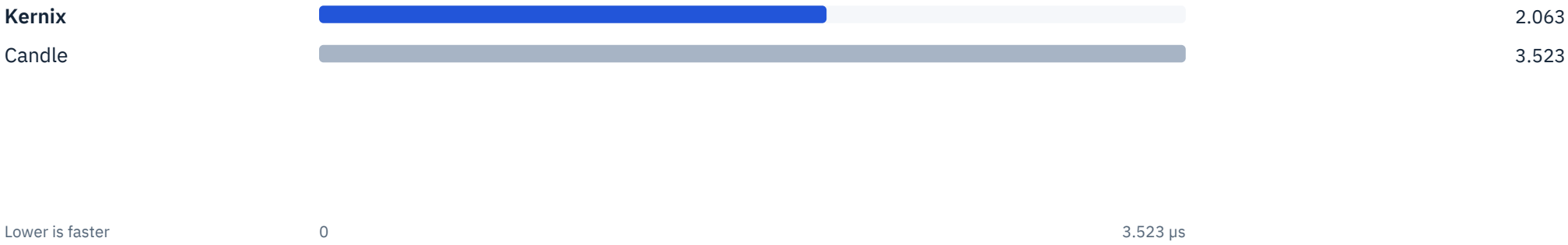
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

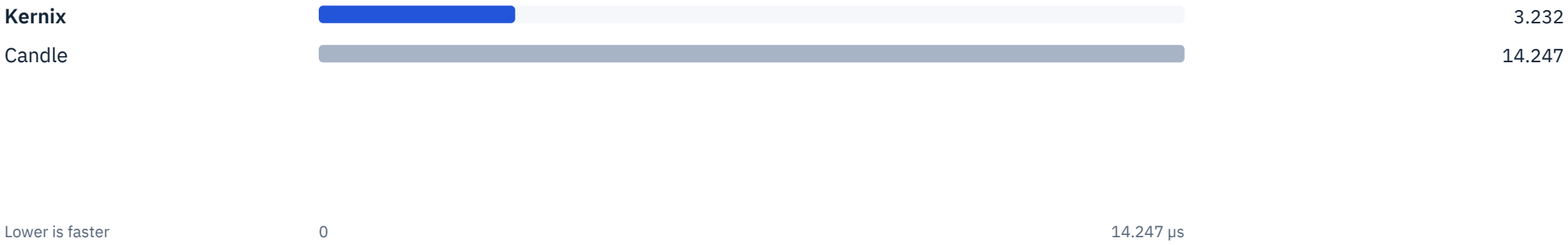
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

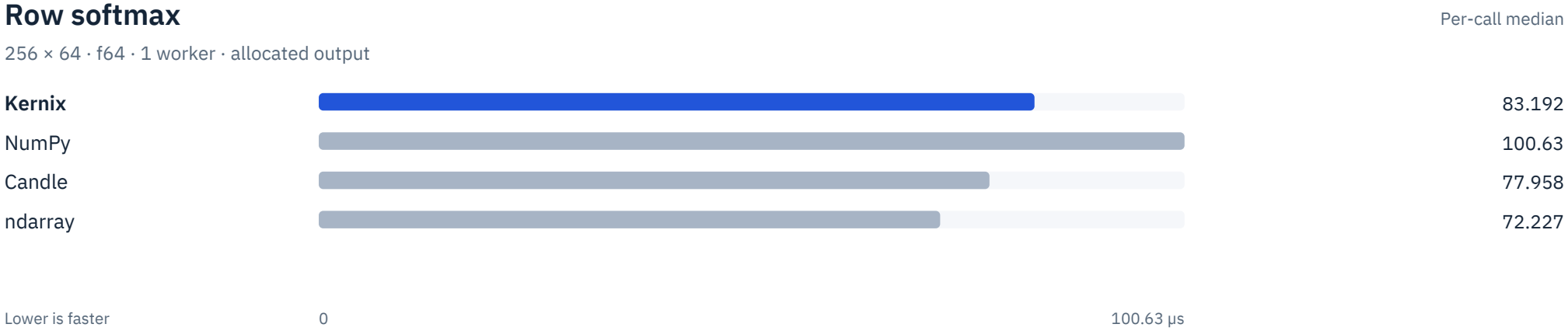


Kernix 0.7.1 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



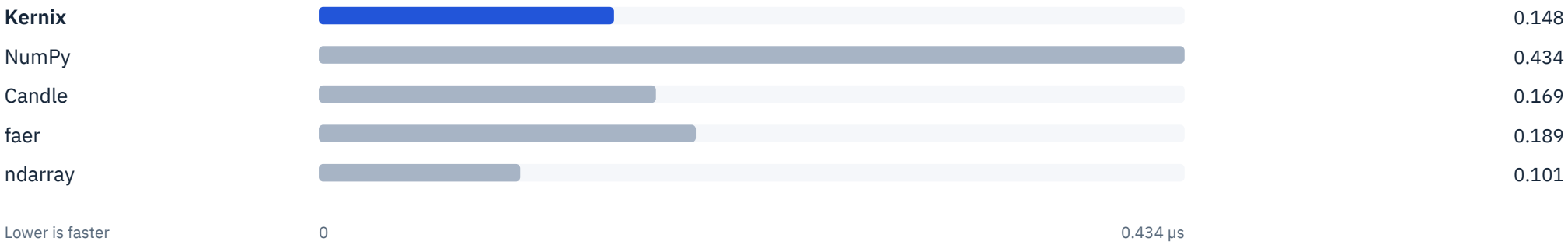
Kernix 0.7.1 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1648	1.648
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1396	1.396
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1529	1.529
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1616	1.616
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1533	1.533
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2813	2.813
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	11918	11.918
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11235	11.235
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11308	11.308
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10951	10.951

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	10010	10.01
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12842	12.842
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	120337	120.337
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	119341	119.341
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	117145	117.145
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	85355	85.355
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	74649	74.649
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	77968	77.968
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	674471	674.471
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	665291	665.291

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	662539	662.539
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	744669	744.669
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	578302	578.302
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	607544	607.544
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1992.9999999999998	1.9929999999999998
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2084	2.084
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3263	3.263
Thin QR / Kernix 96 × 32 · f64	41627	41.627
Thin QR / nalgebra 96 × 32 · f64	36761	36.761
Thin QR / faer 96 × 32 · f64	64548.00000000001	64.54800000000001

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	39043	39.043
Thin QR / Kernix 256 × 64 · f64	495470	495.47
Thin QR / nalgebra 256 × 64 · f64	369394	369.394
Thin QR / faer 256 × 64 · f64	419047	419.047
Thin QR / NumPy 256 × 64 · f64	346813	346.813
Cholesky factorization / Kernix 64 × 64 · f64	22754	22.754
Cholesky factorization / nalgebra 64 × 64 · f64	13762	13.762
Cholesky factorization / faer 64 × 64 · f64	8147	8.147
Cholesky factorization / NumPy 64 × 64 · f64	10397	10.397
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2583	2.583

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4326	4.326
Model inference / Kernix batched normalized/sliced MLP · f64	2063	2.063
Model inference / Candle batched normalized/sliced MLP · f64	3523	3.523
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2563	2.563
Model recording + forward / Candle batched normalized/sliced MLP · f64	5375	5.375
Model backward / Kernix prepared graph → fresh gradients · f64	3232	3.232
Model backward / Candle prepared graph → fresh gradients · f64	14247	14.247
Parameter update / Kernix transactional SGD, fixed gradients · f64	596	0.596
Row softmax / Kernix 256 × 64 · f64	83192	83.192
Row softmax / ndarray 256 × 64 · f64	72227	72.227

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / Candle 256 × 64 · f64	77958	77.958
Row softmax / NumPy 256 × 64 · f64	100630	100.63
Create zeros / Kernix 64 × 64 · f32	148	0.148
Create zeros / ndarray 64 × 64 · f32	101	0.101
Create zeros / faer 64 × 64 · f32	189	0.189
Create zeros / Candle 64 × 64 · f32	169	0.169
Create zeros / NumPy 64 × 64 · f32	434	0.434
Create ones / Kernix 64 × 64 · f32	193	0.193
Create ones / ndarray 64 × 64 · f32	147	0.147
Create ones / faer 64 × 64 · f32	179	0.179

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / Candle 64 × 64 · f32	225	0.225
Create ones / NumPy 64 × 64 · f32	1352	1.352